UNIVERSITY OF BRISTOL

School of Mathematics

FINANCIAL MATHEMATICS 3

MATH35400

(Paper code MATH–35400)

Summer 2025 2 hours 30 minutes

This paper contains 4 questions. All answers will be used for assessment.

- Any calculators are permitted.
- Candidates may bring one hand-written sheet of A4 notes written double-sided into the examination. Candidates must insert this into their answer booklet(s) for collection at the end of the examination.

You may find the following results useful:

- The *put-call-parity* states that

$$S_t + P_t - C_t = Ke^{-r(T-t)}$$

where S_t is the time t price of the asset, $Ke^{-r(T-t)}$ the discounted value of the bond and C_t and P_t the prices of European call and put options at time t.

- *Itô's lemma* for the derivative of a function of an Itô process with stochastic differential $dX_t = b_t dt + \sigma_t dW_t$:

$$df = \left(f_t + b_t f_x + \frac{1}{2}\sigma_t^2 f_{xx} \right) dt + \sigma_t f_x dW_t.$$

Do not turn over until instructed.

1. **(25 marks)**

(a) Consider a financial market with one risk-free asset with price process $B_0 = 1, B_1 = 1.1$ and one risky-asset with price process $S_0 = 100$ and

$$S_1 = \begin{cases} 120, & \text{with probability } 0.5, \\ 60, & \text{with probability } 0.5. \end{cases}$$

i. (**3 marks**) Find a risk-neutral measure for this financial market. Is this financial market free of arbitrage? Is it complete? Justify your reasoning.

ii. (**3 marks**) Find the replicating portfolio of a call option with strike price $K = 90$. What is the premium that you would be willing to pay to purchase this call option at time $t = 0$?

iii. (**2 marks**) Find the price of a put option with the same strike price $K = 90$.

iv. (**2 marks**) Find the replicating portfolio for the contingent claim $X = (S_1)^2$. What is the price of this contingent claim at time $t = 0$?

(b) (**6 marks**) Consider a single-period model with two risky assets that have initial prices $S(0) = (S_1(0), S_2(0)) = (10, 100)$ and time 1 prices

$$S(1) = (S_1(1), S_2(1)) = \begin{cases} (12, 130), & \text{with probability } 0.3, \\ (7, 60), & \text{with probability } 0.7. \end{cases}$$

There is a risk-free bank account with interest rate $r = 0.1$. Decide whether this financial market is free of arbitrage. If yes, calculate a risk-neutral probability measure. If not, find an arbitrage opportunity.

(c) (**9 marks**) A *double call option* can be exercised either at time T_1 with strike price K_1 or at time $T_2 > T_1$ with strike price K_2. Assume the existence of a risk-free bank account with continuous compounding and interest rate r.

Use the no-arbitrage principle to show that it is not optimal to exercise at time T_1 if $K_1 e^{-rT_1} > K_2 e^{-rT_2}$.

Continued...

2. (**25 marks**) Consider a multi-period model with one stock *Ozzy* over three months and price $S_t, t = 0, 1, 2, 3$. There is a bank account with monthly interest rate $r = 0.1$, so that £1 invested on the bank account at time 0 is worth $B_t = 1.1^t$ for $t = 0, 1, 2, 3$.

   Let also P_t and C_t be, respectively, the prices at time t of European put and call options on *Ozzy* maturing at time T with strike price K.

   (a) i. (**2 marks**) Show that

   $$S_t + P_t - C_t = K/1.1^{3-t} \quad \text{for } t = 0, 1, 2, 3.$$

   [Hint: You can use that if two portfolios have the same value at time T, then they must have the same value for all $t < T$.]

   ii. (**2 marks**) The option prices P_0 and C_0 depend on the values of S_0 and K. Which of the following inequalities are true for all S_0 and K and why?

   A. $P_0 \leq S_0$;

   B. $P_0 \leq K$.

   (b) The current price of *Ozzy* stocks is $S_0 = £64$. An investor models the price of *Ozzy* for the next three months and believes that each month the stock either increases by a factor of $u = 1.25$ or decreases by a factor of $d = 0.5$. The interest rate during this period is fixed at $r = 0.1$ per month.

   i. (**2 marks**) What are the possible stock prices S_t for *Ozzy* at times $t = 1, 2$ and 3?

   ii. (**1 mark**) Calculate the risk-neutral probability for moving up for each period in the model.

   iii. (**3 marks**) What is the risk-neutral time 0 price of a European put option with strike price $K = 35$ that matures after three months?

   iv. Consider now an American put option with strike price $K = 35$.

   A. (**2 marks**) Determine the payoff of this option at each state of the model.

   B. (**9 marks**) Calculate the risk-neutral price of this option at time 0. State its optimal exercise strategy.

   C. (**2 marks**) A company selling the option wants to build up a replicating strategy. What would the replicating portfolio be for the first period (no need to calculate the strategy for the second and third period!)?

   D. (**2 marks**) What is the price of an American call option, again with strike price $K = 35$?
   [Hint: You may use any results that you know including the put-call-parity.]

3. (**25 marks**)

Let $\{W_t\}_{t\geq 0}$ be a standard Brownian motion, i.e. $W_0 = 0$, $W_{t+u} - W_t \sim \mathcal{N}(0, u)$ independent of the past values W_s, $s < t$, and $\{W_t\}$ has continuous paths.

(a) i. **(2 marks)** Show that for any $s, t \geq 0$ the covariance satisfies Cov $(W_s, W_t) = \min(s, t)$.

ii. **(5 marks)** Define new processes $\{B_t\}$ and $\{\tilde{B}_t\}$ by

- $B_0 = 0$ and $B_t = tW_{1/t}$ for $t > 0$.
- Let c be a positive constant. Define $\tilde{B}_t = \sqrt{c}W_{t/c}$.

Show that Cov $(B_s, B_t) = \min(s, t)$ and Cov $(\tilde{B}_s, \tilde{B}_t) = \min(s, t)$.

(b) (**12 marks**) Put each of the following processes $\{X_t\}$ in standard form $dX_t = \mu_t dt + \sigma_t dW_t$, determine whether $\{X_t\}$ is a martingale and find $\mathbb{E}(X_t)$ and Var (X_t).

i. $X_t = \exp(\sqrt{2}W_t - t)$

ii. $X_t = t^2 W_t$.

(c) **(6 marks)** Apply Itô's lemma to W_t^4 and calculate $\mathbb{E}(W_t^4)$. Show that for a normally distributed random variable $X \sim \mathcal{N}(0, \sigma^2)$ its fourth moment is $\mathbb{E}(X^4) = 3\sigma^4$.

Continued...

4. (**25 marks**) Consider a Black-Scholes model consisting of a riskless bank account B_t with interest rate r and a risky asset with price process

$$S_t = S_0 \exp\left(\sigma W_t + \left(\mu - \sigma^2/2\right) t\right).$$

The Black-Scholes formula for the price at time t of a European call option with strike price K and maturing at time T is given by

$$C(t) = S_t \Phi(d_1(S_t, T-t)) - Ke^{-r(T-t)}\Phi(d_2(S_t, T-t))$$

where the functions $d_1(s,t)$ and $d_2(s,t)$ are given by

$$d_1(s,t) = \frac{\log(s/K) + (r + \sigma^2/2)t}{\sigma\sqrt{t}}$$

$$d_2(s,t) = \frac{\log(s/K) + (r - \sigma^2/2)t}{\sigma\sqrt{t}}$$

and Φ is the standard normal cumulative distribution function.

(a) (**3 marks**) Find the price of a European put option with strike price 100, expiry date of 1/2 (i.e. 6 months) when $S_0 = 105, r = 0.1$ and $\sigma = 0.3$.

(b) (**2 marks**) Under what market conditions does there exist a unique equivalent martingale measure $\mathbb{Q}$?

(c) (**3 marks**) Assume that a unique equivalent martingale measure $\mathbb{Q}$ exists and is defined by $\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}\left(L_T \mathbf{1}_A\right)$, where

$$L_T = \exp\left(\frac{r-\mu}{\sigma}W_T - \frac{(r-\mu)^2}{2\sigma^2}T\right).$$

Define the process $\overline{W}$ such that it is a standard Brownian motion under $\mathbb{Q}$. Write S_t in terms of $\overline{W}$.

(d) (**2 marks**) Show that under $\mathbb{Q}$,

$$\mathrm{d}S_t = rS_t\mathrm{d}t + \sigma S_t \mathrm{d}\overline{W}_t.$$

(e) (**5 marks**) Denote by $g(t, S_t)$ the price of a European option which pays off $X = g(S_T)$ at time T. By applying Itô's formula to $e^{-rt}g(t, S_t)$, show that g satisfies the Black-Scholes-Merton differential equation

$$g_t + rxg_x + \frac{\sigma^2}{2}x^2 g_{xx} - rg = 0,$$

and state the boundary conditions.

(f) i. (**2 marks**) Consider a digital lookback option, whose payoff at maturity T is a fixed amount K if the underlying price has reached a certain level L during the life of the option, and otherwise the payoff is 0. Write the price of the derivative at time 0 as a conditional expectation.
Hint: The payoff of the derivative at time T is $K\mathbf{1}_{\{\max_{t<T}(S_t)>L\}}$.

ii. (**6 marks**) Derive an expression for the price of this derivative at time 0.
Hint: You may use that $\mathbb{P}(\max_{t<T}(aW_t + bt) \geq c) = \mathcal{N}\left(\frac{-c+bT}{a\sqrt{T}}\right) + e^{\frac{2bc}{a^2}}\mathcal{N}\left(\frac{-c-bT}{a\sqrt{T}}\right)$.

(g) (**2 marks**) State two drawbacks of the Black-Scholes model.

End of examination.

Probabilities of a standard normal distribution

The next tables shows probabilities from a standard normal distribution. Each cell contains the value of $\Phi(x_1 + x_2)$ where x_1 and x_2 are the labels of the corresponding row and column.

Φ	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990